



TRANSPOL

BERCI

JORFELTOC LDA





TEAM MEMBERS



Francisco Araújo
Team Leader

Francisco was the first to show interest and enthusiasm about this project, encouraging all other members to join it. He was responsible for allocating tasks and ensuring that they were all accomplished, also helping all the other members with their respective tasks.



João Nunes
Design Engineer

João was the most acquainted member with CAD software and wanted to deepen his knowledge on this area. He designed our car's prototypes and tested them in the wind tunnel, using CFD 2021, in order to obtain the best performing car possible.



José Azevedo
*Graphic Designer /
Manufacturing Engineer*

José, alongside our partners, decided which materials were the most favorable for the development of the car. He was also in charge of the design of our portfolio and the car painting.



João Maia
Graphics Designer

João, being the member with the best drawing skills, was in charge of the image and personality of our team, creating and managing our social networks, website and designing our uniforms, logo and car colors.



Miguel Dias
Resource Manager

Miguel was in charge of choosing sponsors and strategic partners, in order to obtain the best financing and possible help to achieve our goal. He organized meetings with them, produced appealing material to our partners and handled reports on our monthly activities.



João Alves
*Resource Manager /
Manufacturing Engineer*

Despite his title, Alves was the most versatile member of this team, helping in all areas of this project, from scheduling meetings, researching materials, working with CAD programs and even designing uniforms.

OUR GOAL

As we have chosen to follow our ancestors' dreams, our goal is to win the regional and national competition in order to represent our nation at the world finals. This is a demanding goal but we have been working hard to achieve it .



ABOUT OUR TEAM

OUR NAME

The Adamastor is a mythological creature that sank the ships that tried to bend the Cape of Storms. The courage and boldness of the Portuguese people in facing such a creature has been marked throughout all our history. We decided to base our team, not on the mythological being itself, but on what he led our people to be. This creature sparked in the Portuguese a never-before-seen courage and capacity for overcoming, that made possible the complete domination of the Seas. Our team resembles our people for our courage, being the first team at our school to partake in such a complex competition like this, that requires a lot of dedication, and in the spirit of overcoming, reaching achievements that we would never have imagined before.



TEAM IDENTITY

Since we started our project in this incredible competition, we have created a website and a page on instagram, @teamadamastor, so that all of our followers and supporters can learn more about the team members, the colors, the meaning behind our team's representative symbols and our car.

Website:

<https://adamastorf1team.wixsite.com/overcomealldds>

OUR SLOGAN

OVERCOME ALL ODS

"Overcome all odds" is more than just a slogan, it's our way of thinking, it's how we face this competition, it's everything our team stands for. We decided to choose this slogan because it's the one that best fits our team's ambitions and objectives with this competition, as we are new to this kind of project. However, we won't allow that to dictate our future in this competition.



ABOUT OUR TEAM

LOGO EXPLANATION



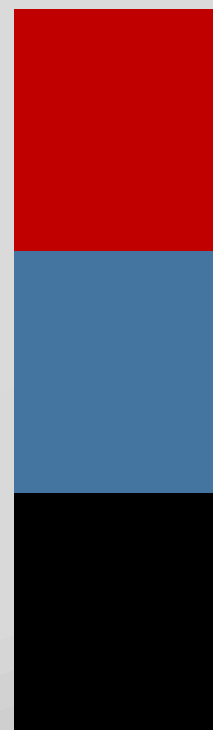
The Half Compass: embraces the upper half of our logo and was a crucial tool for our ancestors, a tool that allowed them to reach their goals. It is, for us, a lucky charm and an indispensable part of our team's representation.

The Helmet: brings out our resilience during our hardest clashes, and the safety with which we execute anything we propose to do, and also represents how "hard-headed" we are when we have an objective in mind.

Team's Name: we decided to include it fully in our logo, so we can highlight who we are in all our external props (t-shirts, masks, polos, etc...).

COLOR SCHEME

Our color scheme was created from the ground up with the intent of representing our ambitions and objectives on this competition, while at the same time being able to pay tribute to our country and its deeds which inspired us as a team. The colors we chose to depict our group were red, black and blue.



The red colour, which is also the predominant color of Portugal's national flag, represents the conquest and joy we strive to achieve as contestants on this competition.

Blue is the most prominent color on our logo and represents our people's connection to the Seas, as Portugal is one of the most sea-bound places, which coincidentally is where our greatest stories and legends as a country come from.

The black represents the unknown, and the hardships we have to endure as we progress further on this competition, and serves as a reminder that no obstacle is too big if we are determined enough to be victorious.

UNIFORMS AND PROPS



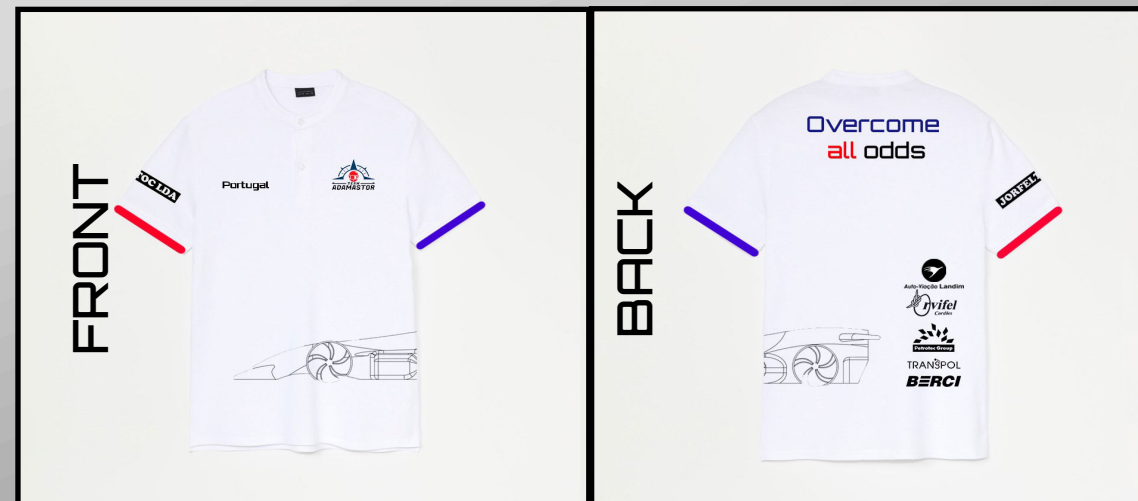
COMPETITION UNIFORMS

Regarding our uniforms, we decided to have two different versions, one for the competition and a more casual one. This is the way we thought about the competition uniform.



For these uniforms, we opted for simple sporty shirts, featuring our logo on the left part of the chest and our country's name on the right. On each arm we featured some of our partners. On the back we decided to place our sponsors in a black tone and our slogan, once more using our team's colors. We chose to put the most important element of this project, our car, on our uniforms since this will be the main reason of our success. This type of shirts were chosen due to their comfort, ease of washing and lightness.

CASUAL UNIFORMS



Regarding the casual uniforms, the white short-sleeved polo couldn't be left out. It is an elegant, sober and very comfortable outfit. We decided to keep the customization plan the same, changing only a few details about the coloration. We opted for the full white polo, and, on each arm, strips dyed with our team's characteristic colors. This way we kept our colors present and created a slightly more formal uniform for meetings with possible partners.

CAR IDENTITY

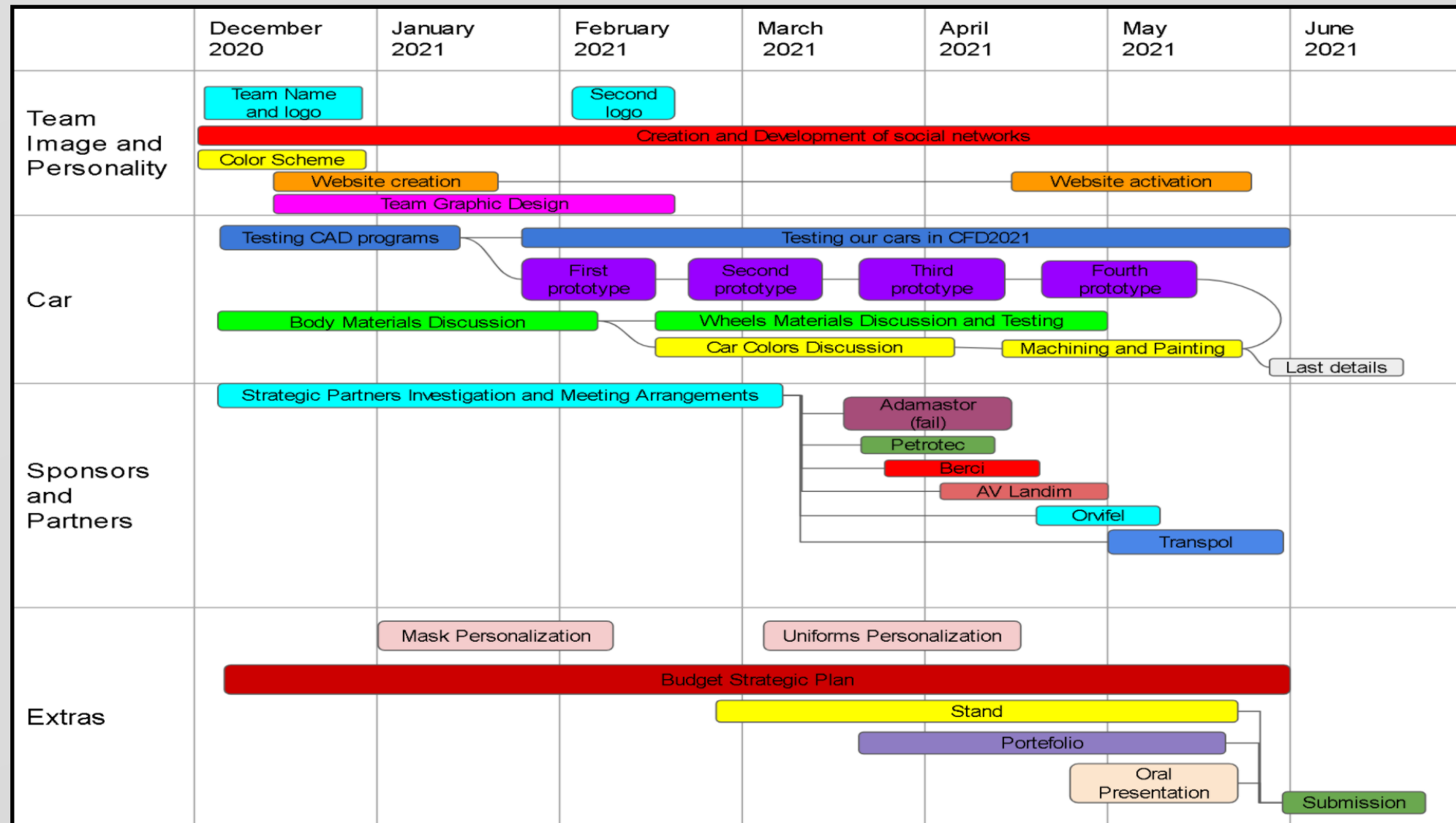
Together with all our exterior props, our car also represents the team, reflecting our colors. The car features a white body, allied with our team's colors on the wheels, the rear airbag, the front airfoil and the upper body, conveying simplicity and beauty, all in once.

OUR PROPS

Our team, alongside some of our partners, opted for the creation of props, in order to capture the public's attention to know and to support our project and to reward our partners and sponsors. We developed a prototype of masks featuring our logo, to promote our team and to appeal to our supporters to be cautious, considering the pandemic we're currently facing.



TIME MANGEMENT



The various segments of our project were allocated to different team members, according to their preference and skills. We started the Team Adamastor project by creating platforms that would provide us with visibility, by familiarizing ourselves with the software needed for the development of the car and by researching the most favorable materials for its accomplishment.

DISCORD AND ZOOM

As it wasn't always possible to bring together all the team members, due to the pandemic we are experiencing, Discord was the most used platform throughout the project. This made it possible to communicate at all times and quickly share all the work done by each member. Zoom was used in most of our meetings with strategic sponsors and made it possible to have "closer" contact with our partners.

Time management was undoubtedly our biggest challenge throughout the project. We have always been used to following orders. Organizing our own ideas and executing them was hard, but we managed to get it done.

In order to guarantee the best possible use of our time, we started by writing down every step we took along this journey. This way it was possible for us to analyze our route, manage our time and make sure we didn't forget about any detail.

SPONSORS/PARTNERS



Regarding our sponsors, we chose to distribute them through different levels, with the intent of enlightening the benefits of each one and organizing our own work, manufacturing all the props taking into account the presence, or not, of each sponsor in each prop.

Being science students, we decided to divide the levels of the sponsors taking into account the value of certain metals: gold, silver and bronze. The gold ones correspond to the highest level and the bronze ones to the lowest level.

Bronze(100€): Advertising in the portfolio, social networks, website and stand.

Silver(200€): Advertising on the jersey and on the perks marked on the level below.

Gold(300€): Advertising on the car and on the perks marked on the levels below.



Auto-Viação Landim

Landim was very excited when we first presented the project to them, and since none of their services were favorable, they contributed monetarily to the project.

Gold

BERCI

This advertising company played a crucial role in the development of this project. Although they do not contribute monetarily, they participated directly in the personalization of our stand and polo shirt, and also in the car stickers.

Gold



Orvifel had a similar role to Landim, monetarily sponsoring our team and sharing contacts that could help us in the future.

Gold

TRANSPOL

Transpol sponsored this project both monetarily and through their services. This company provided us with the various materials needed for the machining of the rear flaps and wheels.

Gold



Petrotec Group was our first sponsor. This company helped us in the machining and design of the car. They were the ones that helped us the most in terms of communication and organization.

Gold

JORFELTOC LDA

Jorfeltoc helped the team monetarily and, together with some of the members, designed our props and uniforms.

Silver



BUDGET MANAGMENT

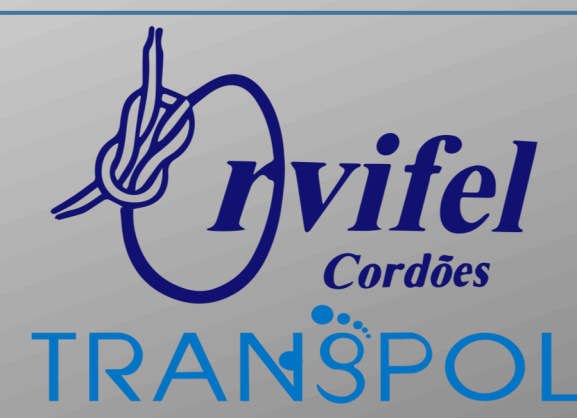
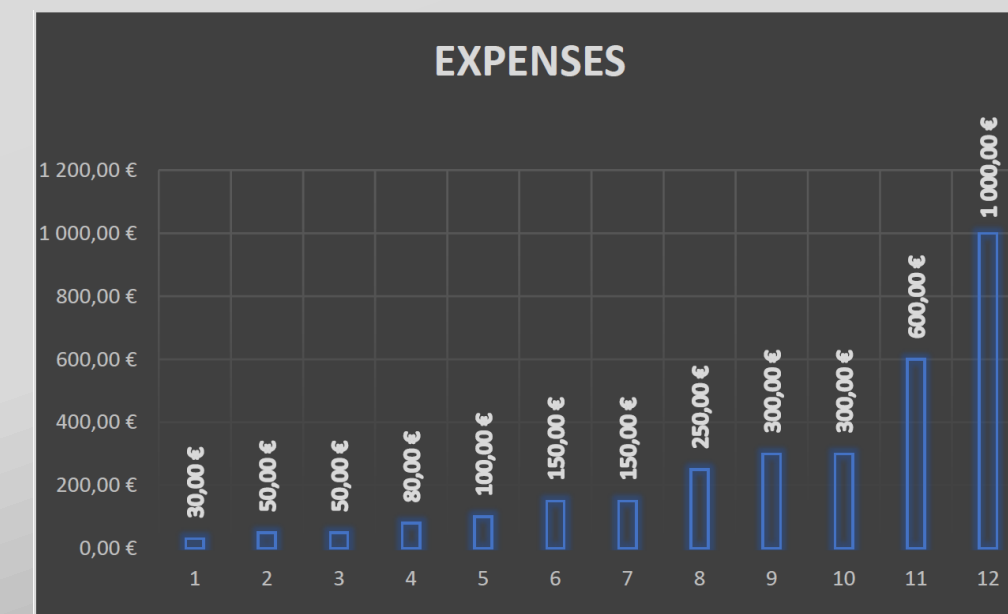
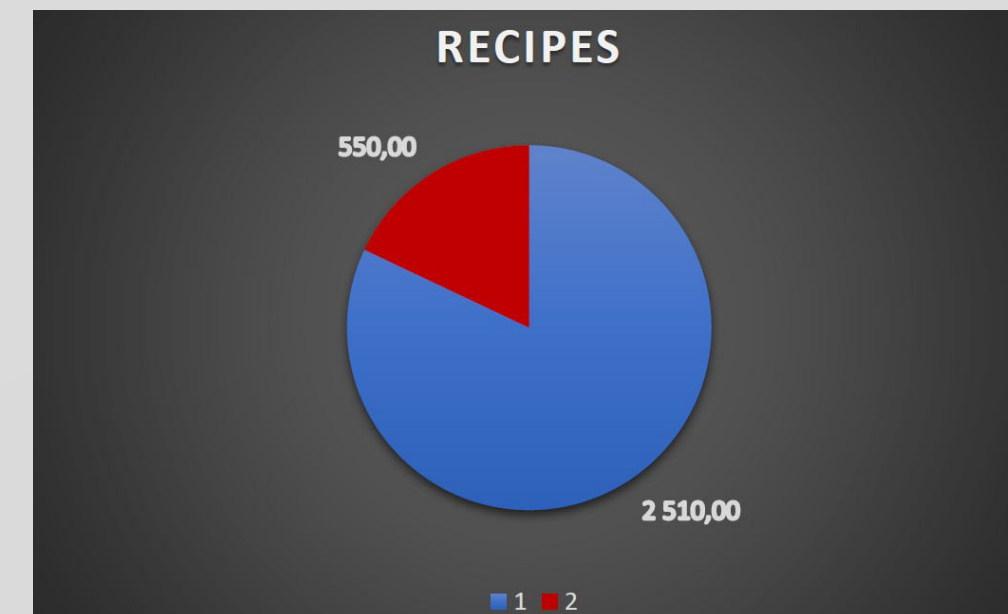
The team's funds were managed in the most cautious way possible. Since our entire project was created and developed during a gigantic pandemic, which brought an economic gap all around the world, we chose to lower the value of the levels related to partnerships and work with the least possible funds. This decision provided a change in our initial projections, making us manage our funds better, and made it easier for companies to support the cause through their services instead of monetarily.

On the right you can see how we chose to manage our budget. Jorfeltoc was the company that helped us the most in terms of budget management. We chose to separate the funds into two groups: the funds from sponsors and the team's own expenses. These groups are distinguished by the colors blue and red, respectively.

Budget	Expenses	Value
1	Polos	30,00 €
2	Car stickers	50,00 €
3	Wheels Machining	50,00 €
4	Masks Costumization	80,00 €
5	Wheels Material	100,00 €
6	Polos Costumization	150,00 €
7	Car Painting	150,00 €
8	Deslocations	250,00 €
9	Consumption and general expenses	300,00 €
10	Ceramic Bearings	300,00 €
11	Stand Costumization	600,00 €
12	Stand Construction (Prediction)	1 000,00 €
	Total	3 060,00 €

Budget	Recipes	
1	Sponsorship	2 510,00€
2	Team's own expenses	550,00 €
	Total	3 060,00€

Own Financing	17,97%
Sponsors Financing	82,03%
Indebtedness	0,00%



STAND AND SOFTWARE



PIT DISPLAY



Our stand was designed with the intent of being visually appealing and clear to everyone. We opted for the placement of two monitors, one with a presentation video of our team, for the public to get to know us better and a second interactive one with our portfolio and with the 3D model of our car.

The stand is personalized with only crucial information related to our project. We decided to place an elevation inside the stand to showcase a prototype of our car on display. We also opted for a 3D printing of a prototype with triple the measures, to allow for a closer contact of our fans with our car.

SOFTWARE USED

The CFD2021 was a crucial software during the development of our car. It was used to aerodynamically test all the parts of our car.



Using Autodesk Fusion 360 we designed all the prototypes of our car and our stand, in a 3D version. The software always provides us with a great final result.



Adobe Photoshop was mostly used to design the team's visual identity and all the external props, like the polo shirts, t-shirts and masks.



Microsoft Publisher was our first choice when it comes to the creation of our portfolio, due to the number of unique tools that enable a high quality final result.



THE BEGINNING

There are a lot of forces that may influence the performance of a F1 car. Some of these either propel the car or slow it down. We started designing our prototype taking into account the following:

- The thrust (which propels the car);
- The drag force (which slows down the car);
- Lift or Down Force;

Thrust

In our case, the thrust is not something we can control, as it is caused by the expulsion of CO₂, coming from the CO₂ canister. The only way we can get the most out of this force is to decrease the car's weight, which results in greater acceleration and, consequently, greater terminal speed, and directing it towards the car's center of gravity.

Drag Force

This force acts along the car, presenting the same way and direction as the air flow. This decreases the speed of the car, as it's contrary to its movement. This force is a combination of several, including air resistance and rolling resistance.

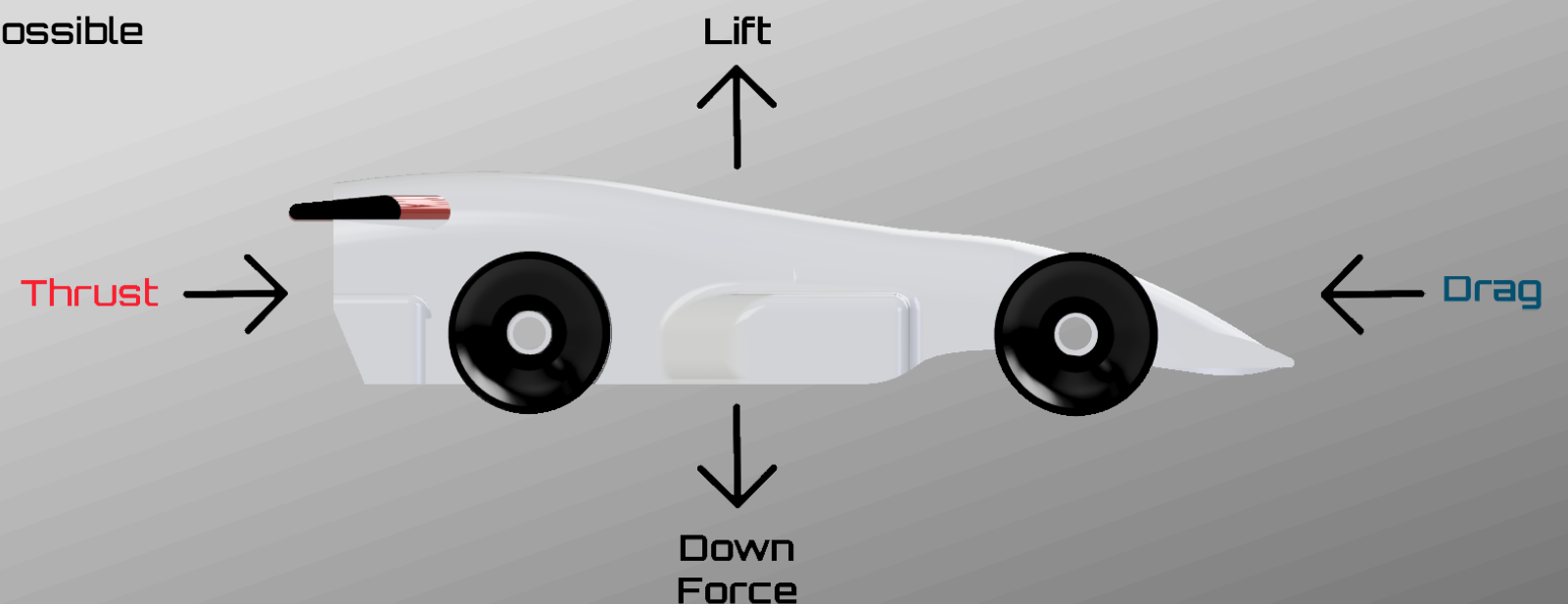
- Air resistance: This force is influenced by the air density, something uncontrollable and, theoretically, equal for all teams in the competition, and the car's aerodynamics, something totally manageable by us and, obviously, thought out and worked on, in order to obtain the lowest possible coefficient of drag.

- Rolling Resistance: The Rolling resistance depends on the car's weight, bearings and friction between the wheels and the track. We opted for ceramic bearings for the lowest possible bearing strength. For the wheels, we have tested several materials, from Polypropylene, ASA filament to 30% carbon fiber filament

Lift or Down Force

The lifting force acts along the vertical axis of the car. This can either make the car go up or increase the Down Force, which pulls the car down.

These forces depend on the aerodynamics of the car, both of which can be beneficial or not, depending on what we want from our vehicle. We based our final prototype on a aeronautical design, thinking of our prototype as an "aircraft with wheels". Therefore, our car is designed to stay mostly off of the ground at the beginning of the race, gaining the highest possible initial speed. In other words, we bet on the lifting force to keep it in the air as long as possible.



INNOVATION AND EVOLUTION

Car Innovation

Regarding the car's design, we developed a side-pov that allowed us to transport the air that passed over the front wheels to the bottom of the rear wheels. This project, unlike our final prototype, was based on down force. However, despite all our efforts to make this mechanism a worldwide evolution, this was not possible, as CNC machines would not be able to machine the air inlet and outlet. This impossibility of machining alongside the new ideas of betting on lift force, coming from the team members, led us to give up on this revolutionary projection.

We also developed axes with built-in suspension, made from aluminum due to the ease of handling, lightness, strength and price. However, we didn't choose to use these due to their uselessness in the racing context. These would be slightly heavier than normal axes and, as during the race we chose to keep the car mostly in the air, this would not be beneficial. .

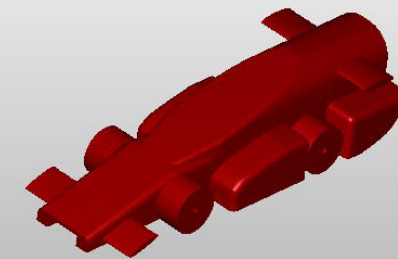


Team Innovation

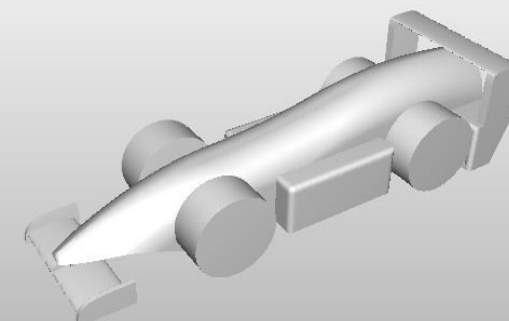
Since the beginning of this project, we have strove to be unique and leave our mark in the competition's history. We started right away by being the only team in our school to join this competition.



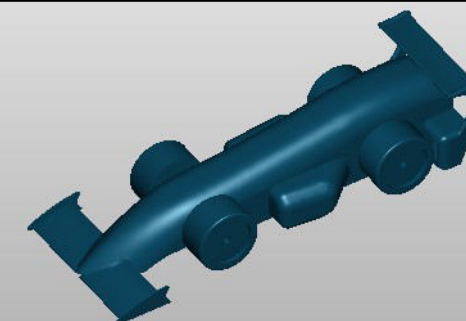
Car Evolution



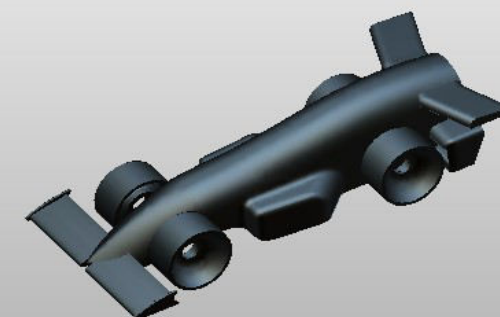
MARK1



MARK2



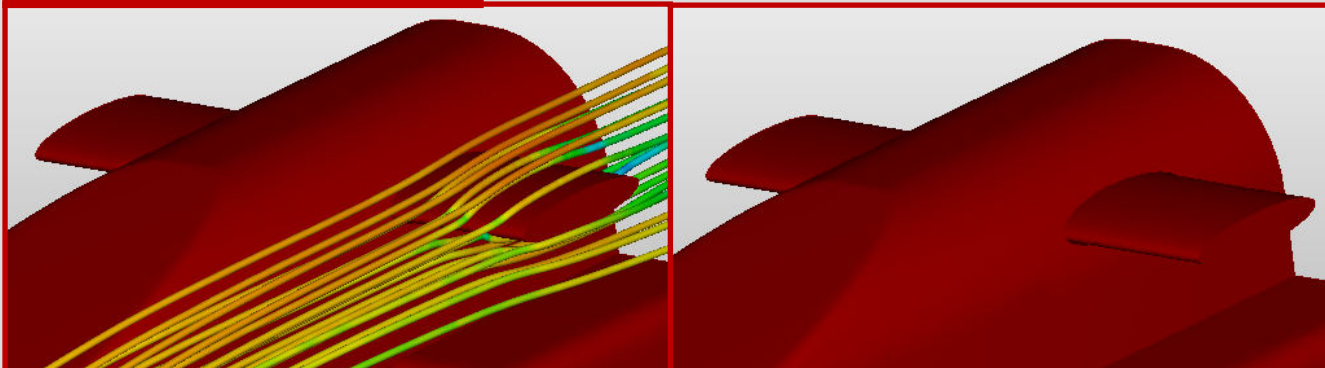
MARK3



MARK4

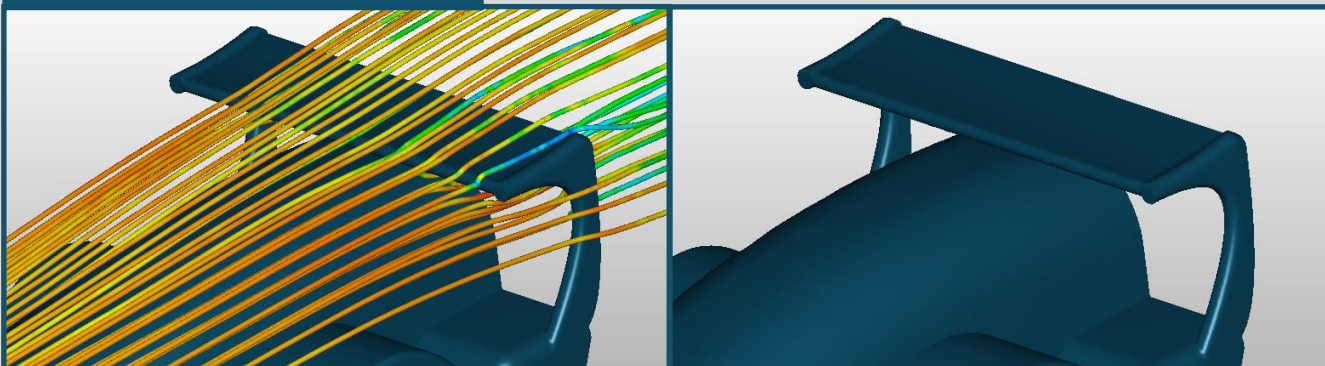
REAR WING

MARK1



It is designed so that the wind is cut by the wings, enhancing some support as the air passing over the wing needs to travel a greater distance when compared to the air passing under the wing, thus creating greater pressure below of the wing providing lift. The ideas presented here served as a launch pad for our final prototype.

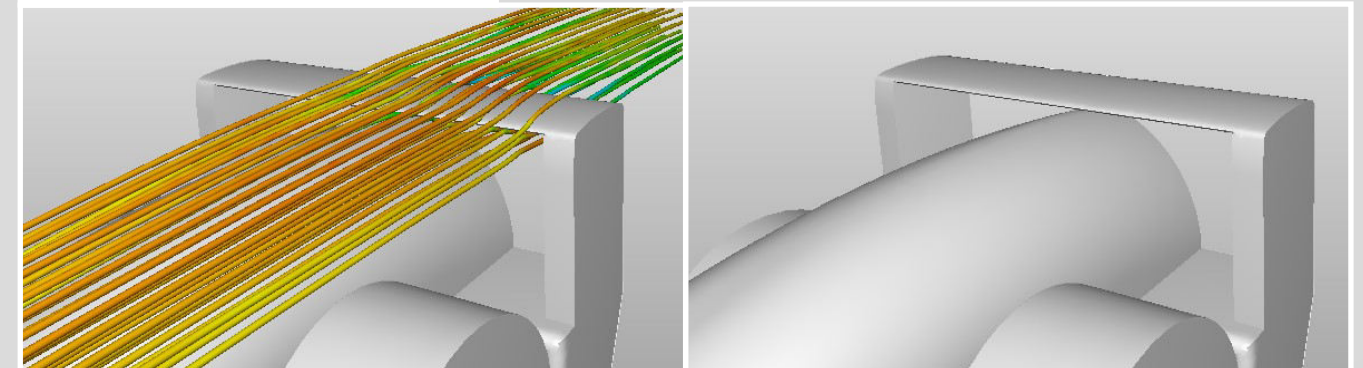
MARK3



Here we have the evolution of the previous design keeping the same principle. We wanted pressure to be generated at the top of the wing so that the car could go "glued" to the ground.

The difference between this and the previous prototype is just the thickness of the wing. We opted for a thinner wing to reduce turbulence.

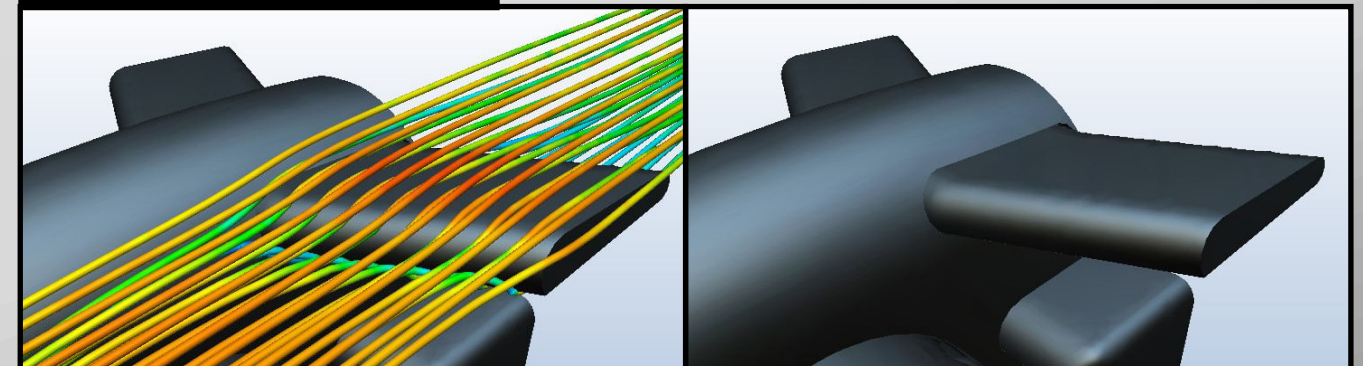
MARK2



In this wing prototype we changed the approach and influenced by the design of the formula 1 cars, we wanted the highest possible downforce.

As we could see, the lines passing through the wing have a similar color and inflow both above and below the wing, which means that the air travels the same distance on both sides, resulting in equal pressure.

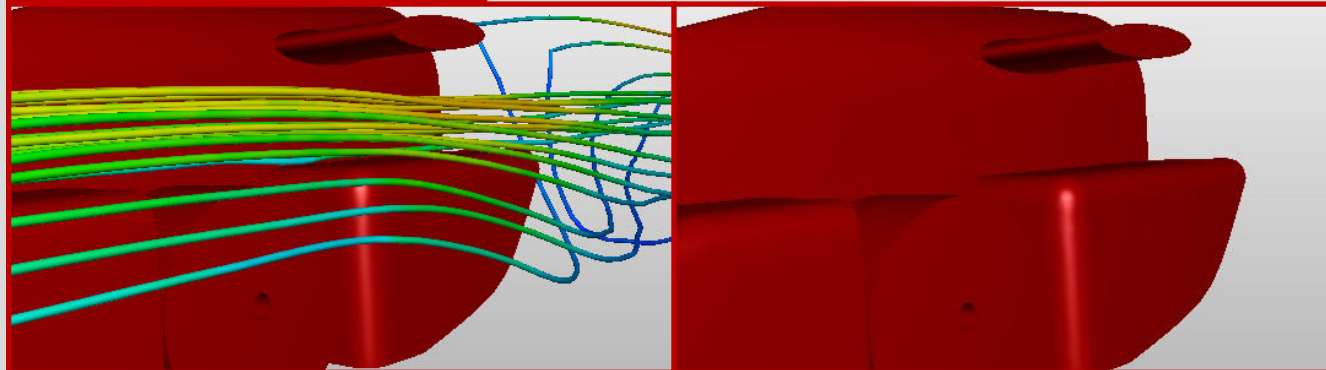
MARK4



In our last wing prototype and with a lot more knowledge about aerodynamics, we noticed that the concept of the first prototype could be reused. Now based on aeronautics, we've inflated the wings and strived for the car to have the perfect balance between lift and downforce.

REAR SIDE-POV

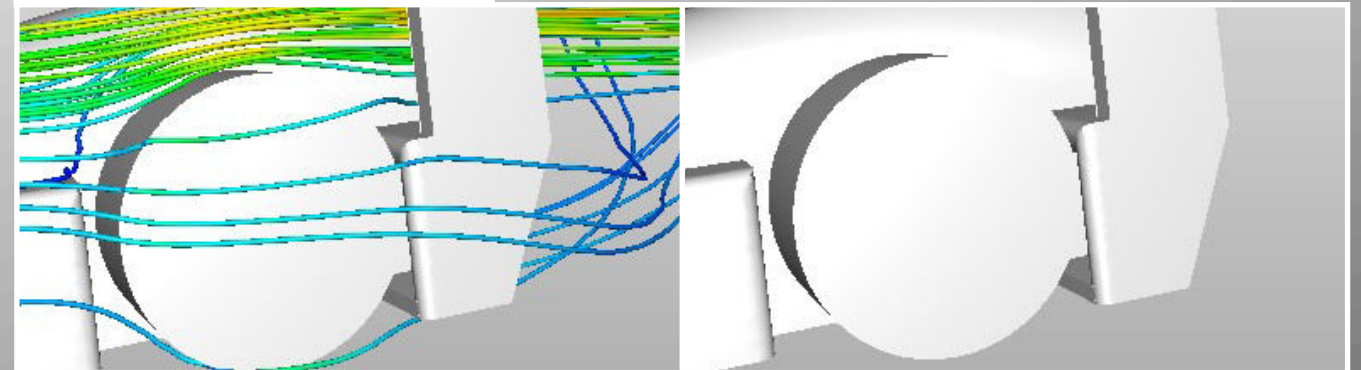
MARK1



The rear side-pov is not the element that most influences the car in terms of speed. This has the function of giving some stability to the prototype.

Our first design of this element was based solely on our intuition and was for testing purposes only.

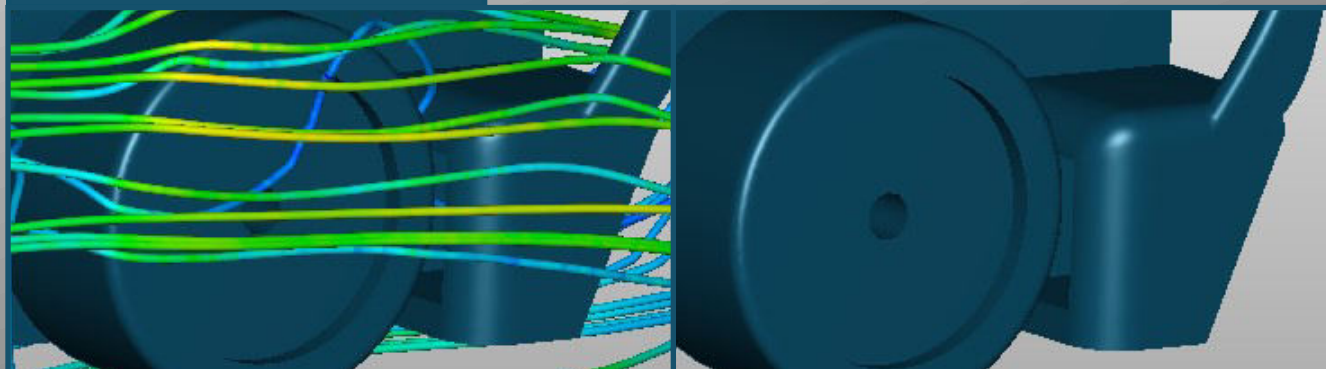
MARK2



After some research, we chose to join the rear part to the airfoil, which resulted in greater rear stability of the car. This was a good result considering the airfoils we had.

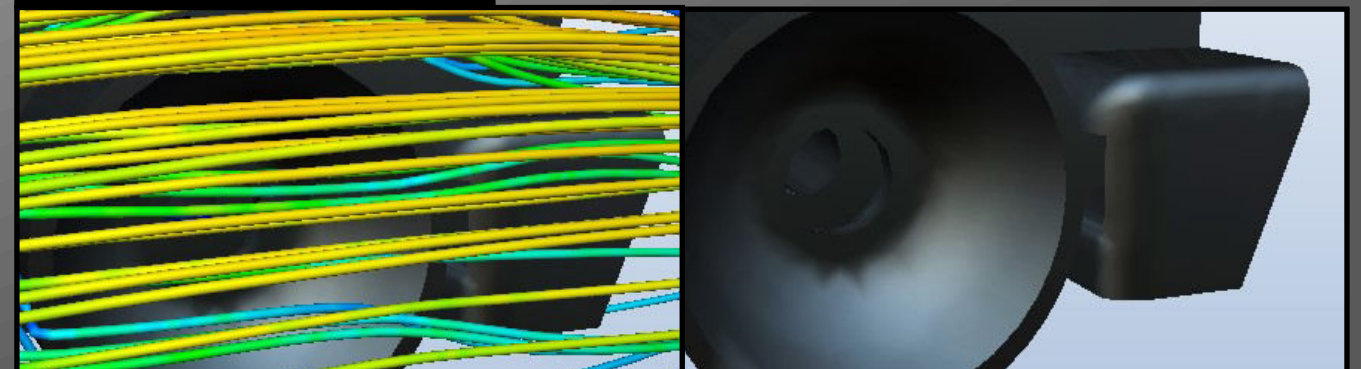
The airfoil attached better to the car's body, however the car was overweight..

MARK3



Taking into account the last test results, we decided to choose to focus on weight reduction. Therefore, together with a reduction in the thickness of the wing, we continued with the opening of our rear side-pov, maintaining the stability of the car and reducing its mass. This design continued in line with our down force-focused ideologies.

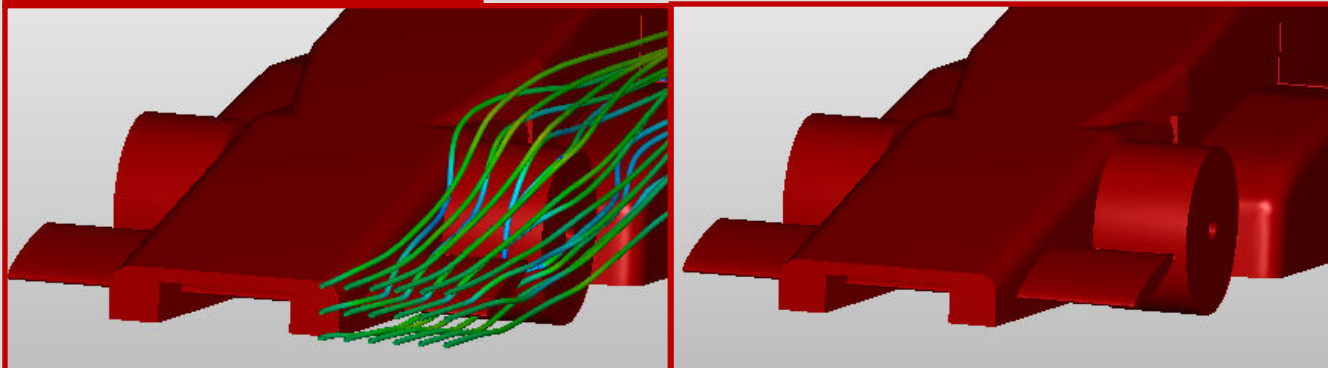
MARK4



In the last prototype, due to a change in the bet on the down force for the balance between this and the lift, the wings were changed. However these did not need to be attached to the rear side-pov to remain stable and well fixed. Therefore, we separated them again. The side-pov gave the car pretty good stability, so we decided to keep this prototype.

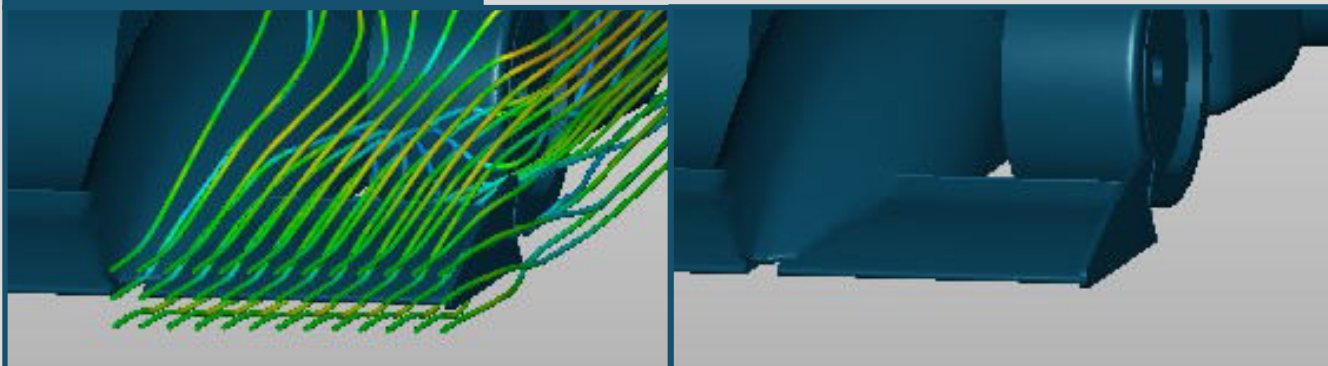
FRONTAL WING

MARK1



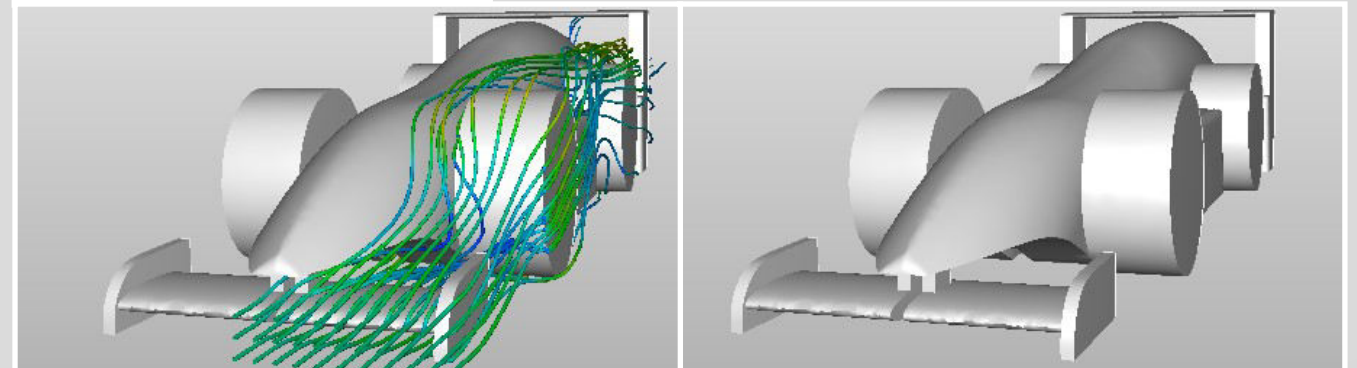
In our first design we focused primarily on the simplicity and effectiveness of our prototype's front wing. We used an ellipse shape with a very low drag coefficient so that while we were gaining down force we didn't lose too much speed due to drag.

MARK3



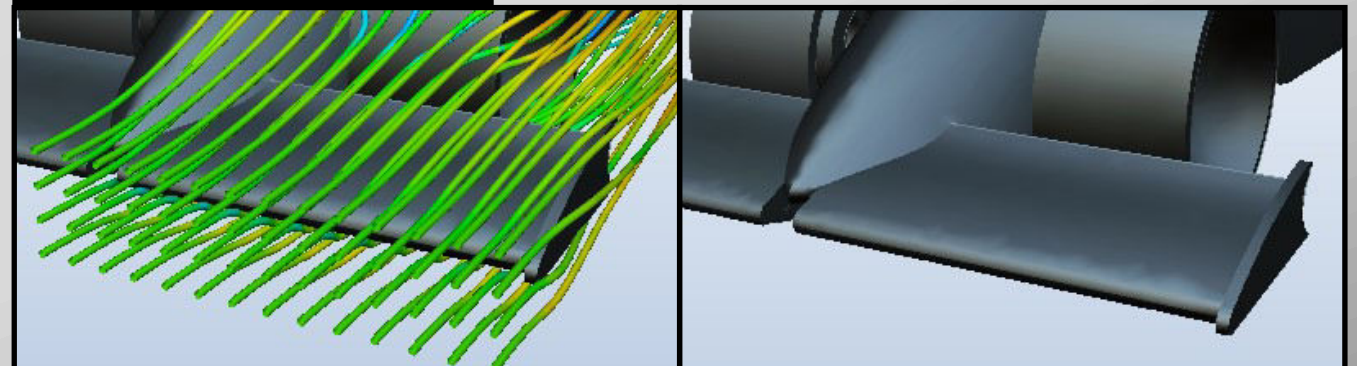
After some research we developed a bolder design, not forgetting the first principle established. For that we lowered the car's nose again, but this time with a spigot design. Regarding the wings, we were inspired by well-known models of formula 1 cars, trying to make the air movement as uniform as possible, thus gaining more down force in the front.

MARK2



In our second prototype we developed a new wing design, keeping the same principle as the first, but this time with a higher nose and, consequently, with the fitting underneath. However, when we lift the nose of the car we would be losing down force, completely discarding this design.

MARK4

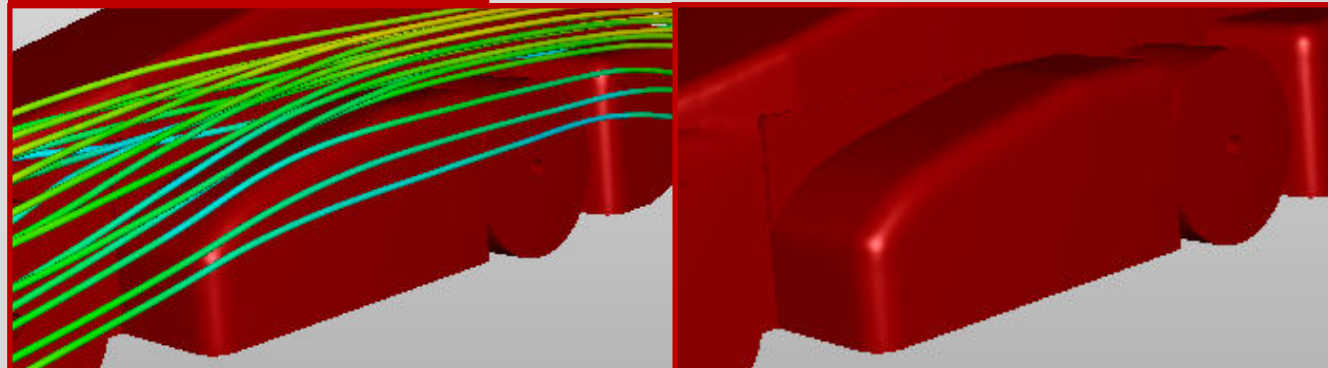


In our last and final design, we chose to use the same wing shape but, compared to the last one, we shortened the nose thus bringing the wing closer to the wheel, not causing as much resistance to the wheel as in the previous model.

SIDE-POV

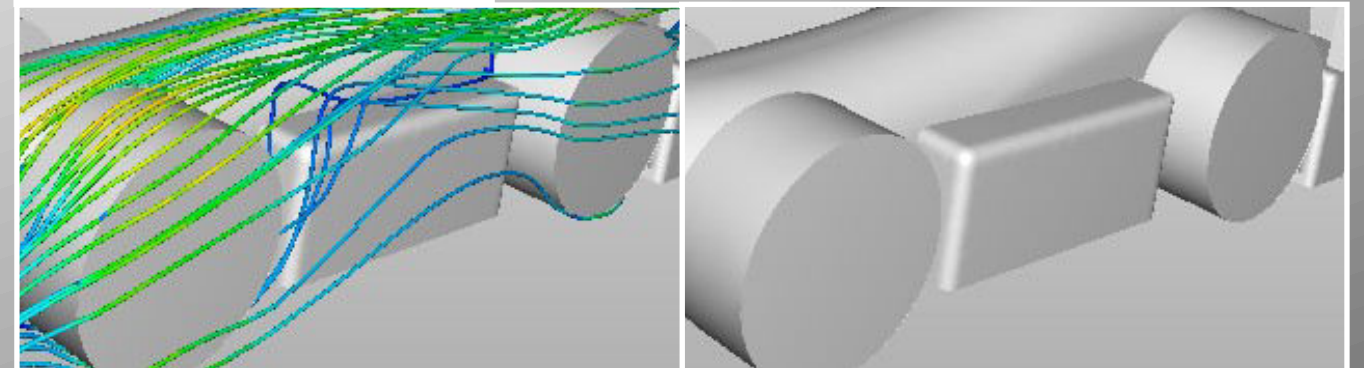


MARK1



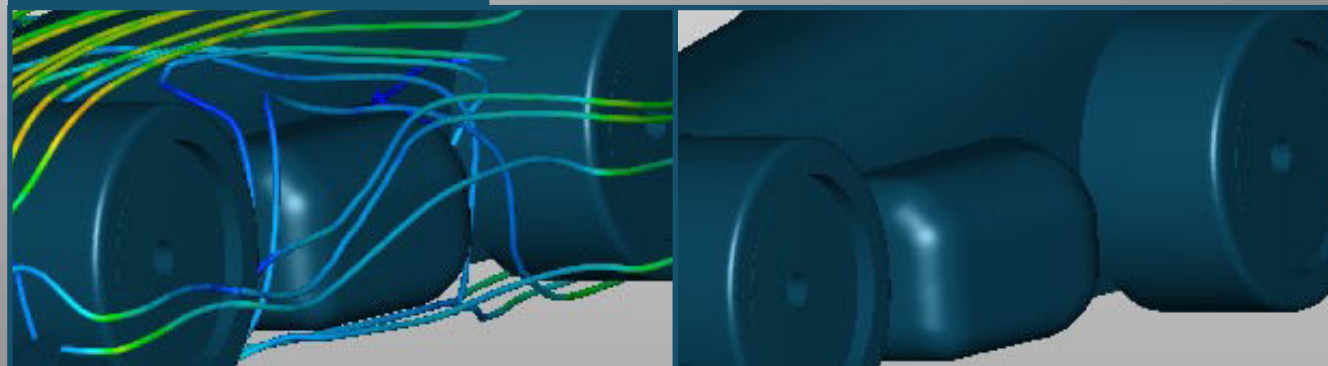
This version of the side-pov was pretty rudimentary but served its purpose, which was to lift the incoming wind away from the back wheels in order to get the most out of them. Despite being later changed we kept the original formula in mind.

MARK2



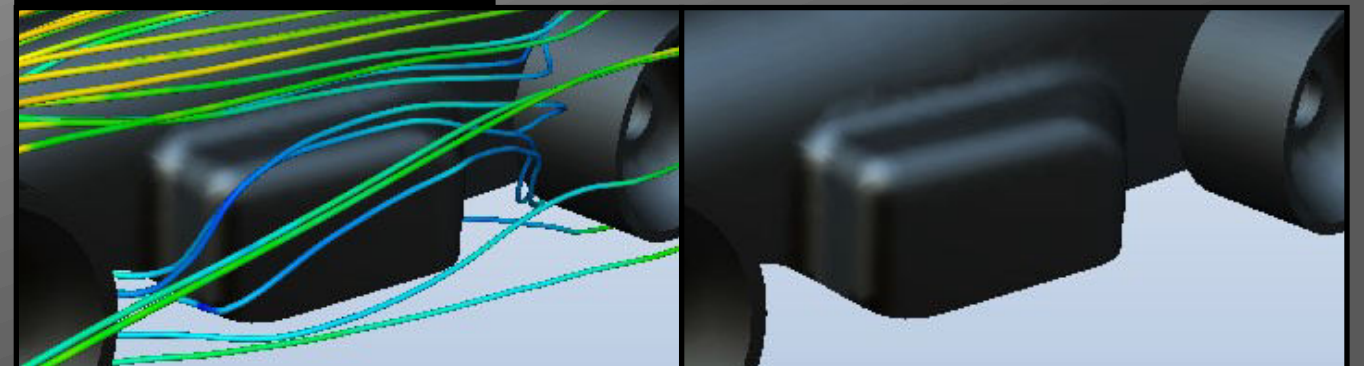
During this fase of the competition we were still learning how to properly use the CAD software and experimenting with different body designs, so the efficiency of the side-pov was greatly reduced, that's when we decided to completely rethink the design the next time around.

MARK3



By the time we designed the third version of the side-pov, the car's body shape was already decided and we had already mastered the CAD software, so it allowed us to further improve its design. The main changes were the reduced height and size, and the rounded corners, that allowed the airflow to circumvent the back wheels a little more easily, but enough to get the most out of them.

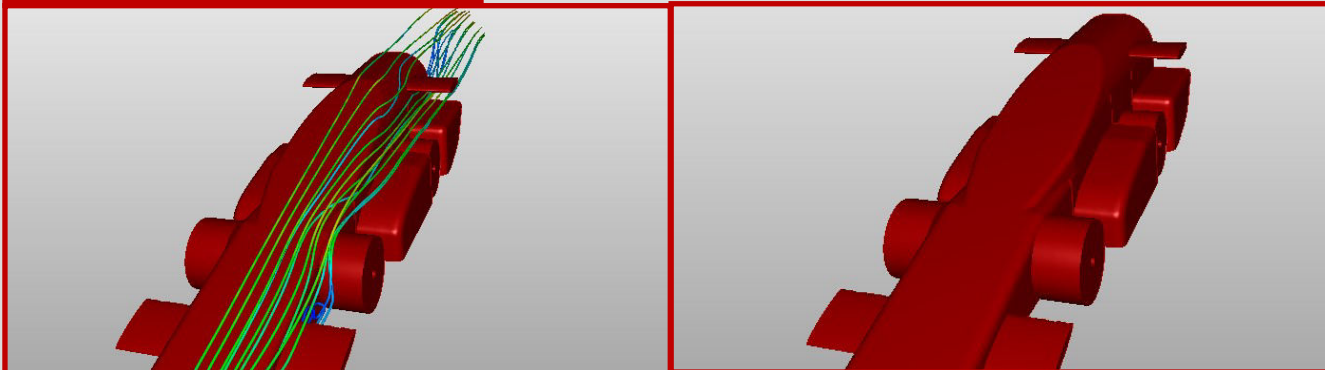
MARK4



This is the final and best take on the side-pov so far, we developed it along with the body of the car in order to create the best possible aerodynamics with all the gathered knowledge so far, its lenght was slightly increased and the corners where further rounded, these changes allowed us to get the best result we could.

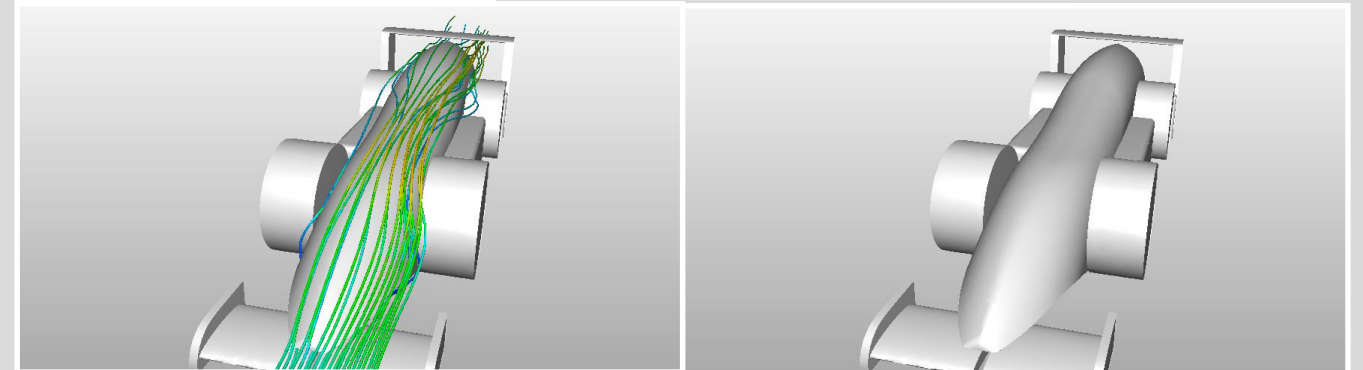
CAR BODY

MARK1



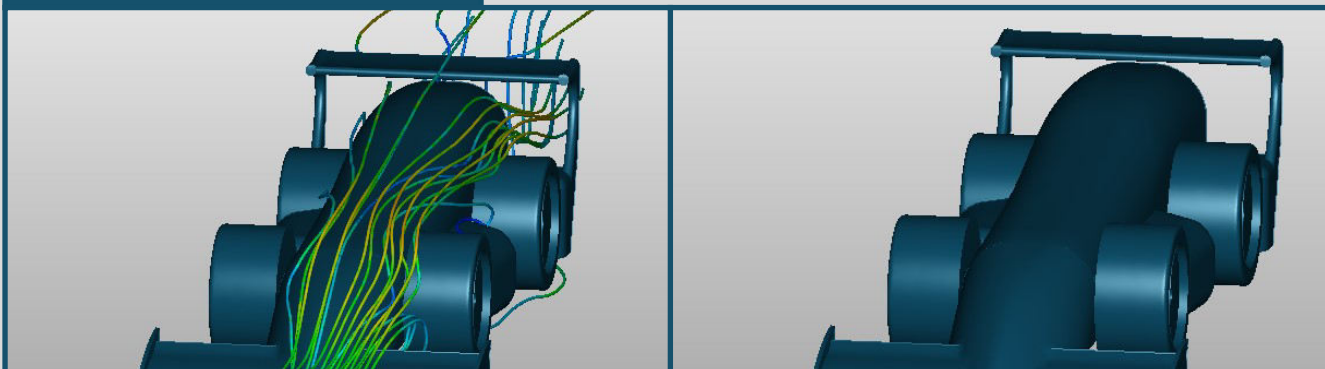
In the first prototype, the experience working with the cad programs was proportional to our knowledge of aerodynamics. This set of factors resulted in a car with too high a drag coefficient for our goals.

MARK2



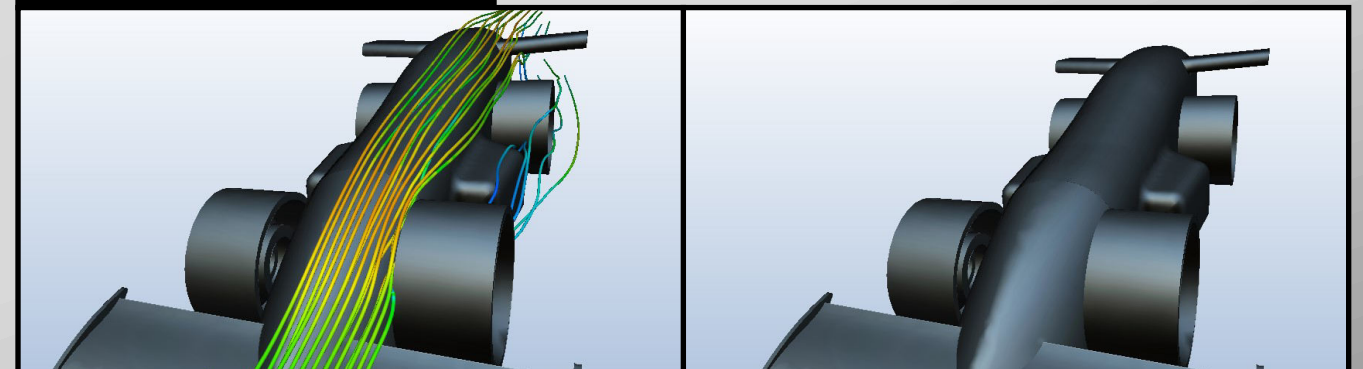
In the second prototype the car body followed the trend of the wings and was redesigned. Despite showing significant improvements the car proved to be quite unstable in the wind tunnel. In the second prototype the car body followed the trend of the wings and was redesigned. Despite showing significant improvements the car proved to be quite unstable in the wind tunnel.

MARK3



In the third prototype we tried to improve the previous one. We redesigned the wings but the body of the car was practically unchanged.

MARK4



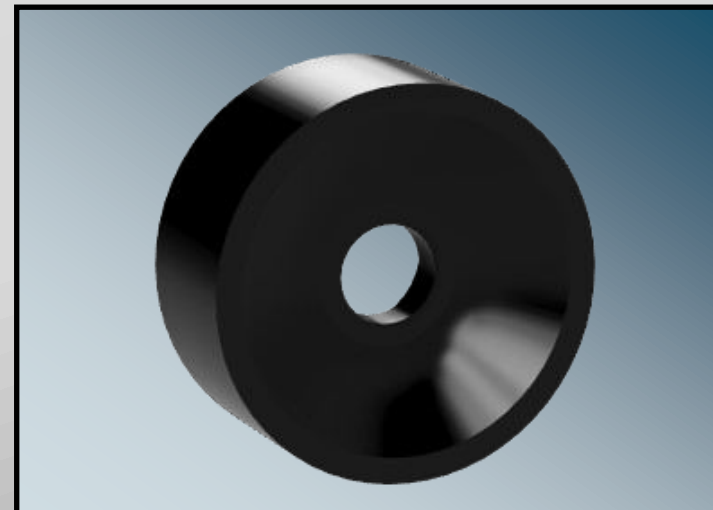
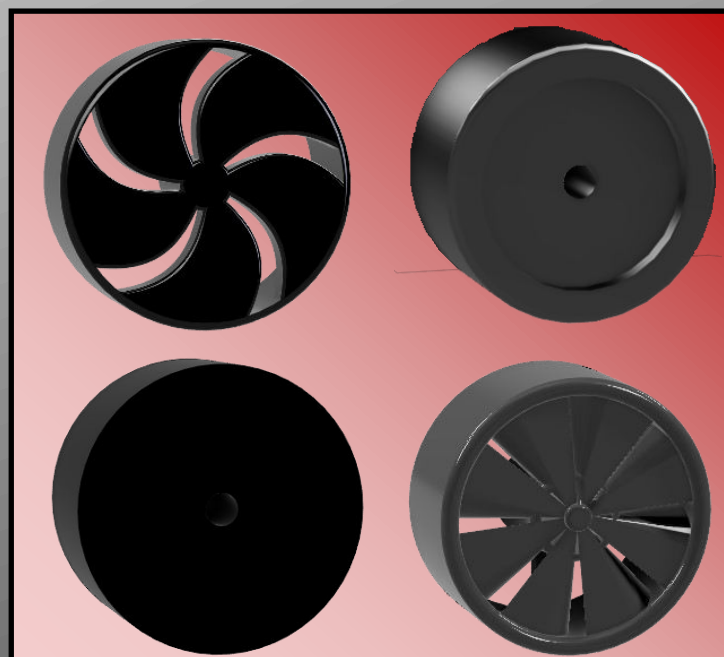
In the fourth prototype the ideology and approach changed. There was a decrease in the height, width and length of the car's body. Turning out to be the most effective configuration for our wing and wheel system. With this body the car revealed a lot of stability in the wind tunnel especially at high speeds.

WHEELS AND MATERIALS

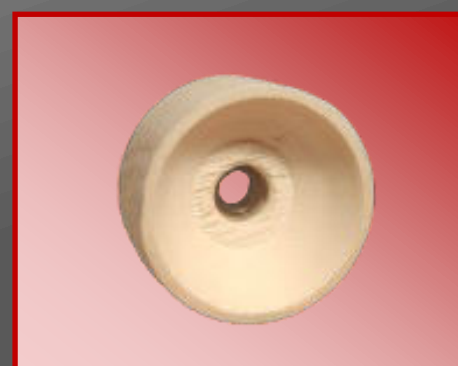
Wheels

Wheels were an element that, like every other part of our car, evolved over the course of the project. However, since the design of the wheels does not influence the prototype aerodynamically, we sought to change the design aesthetically, in order to present unique wheels.

Here, **in red**, are the wheels that we designed and, **in blue**, our final wheels, also designed for us.



Our last wheels weren't the ones that aroused the most interest in visual terms, however, the difficulty of machining was, once again, an aspect that led us to give up on our dream projects. This last prototype, despite not being the most eye-catching in aesthetic terms, conveys some curiosity due to the fact that it has a concave surface, something not very present in this competition.



Materials

For our car, we always try to use the best products within our reach, light, resistant and easy to handle products. For the car body, we use the material provided by the project organization. After a long search of materials, we found several that satisfied our interests, ASA filament, Polypropylene and 30% carbon fiber filament. For our airfoils, both the front and the rear, we opted for the use of carbon filament, as this is more resistant and, despite being heavier, compensates in comparison with the others. For the axes, we opted for aluminum. For our bearings, we chose the ceramic ones.

The material for the wheels was the easiest to choose as, after testing the car on the track, we were soon able to exclude some of the materials in mind, taking **ASA Filament** and **Carbon Filament** as final choices. These will only be chosen on the day of the race as they provide indistinguishable race times.



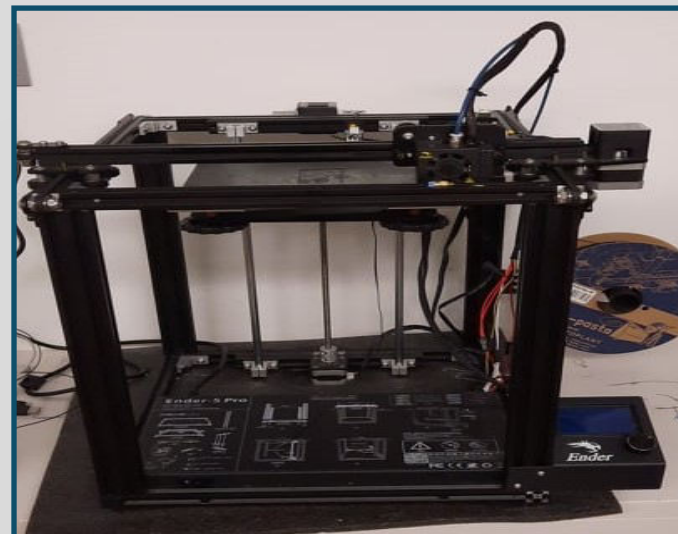
Production

Throughout our project, different means of production were used for each part of the car. For the machining of the body, we chose to use **5-axis CNC machines**, as they allow us to sculpt all kinds of drawings and projections.

Regarding the rear and front airfoils, we opted for 3D printing, through an **Ender 5 Pro printer**, as this type of printing is very fast, very flexible and easy to work with.

The car's axes were made using a **mechanical lathe**, a very versatile machine in the transformation of materials such as aluminum, the material used for these parts.

Both the car body and airfoils were designed using CAD software and later converted to STL files compatible with the 3D printer and CNC machine.



Painting

The process of painting our car was one of the most difficult due to our lack of knowledge in this area. After a lot of research about which types of paints to use and their weight, we came to the conclusion that we should opt for paint that is not too dense, in order to get the less possible painting weight. Before we started painting, we chose to sand the whole car, making the paint reflect a professional look. We start by applying a primary ink to make the surface smooth. Then we apply just one layer to the car. After these steps are completed, we place the stickers. Only at the end do we put an adhesive layer all over the car. Taking into account the possibilities at our disposal for painting, we decided to use car paint in order to make our car visually appealing and to keep its low weight.

FINAL CAR



Label

1- Front Airfoil:

The front airfoil is designed for the air to rise above the car. It was printed using a 30% carbon fiber filament.

2- Wheels:

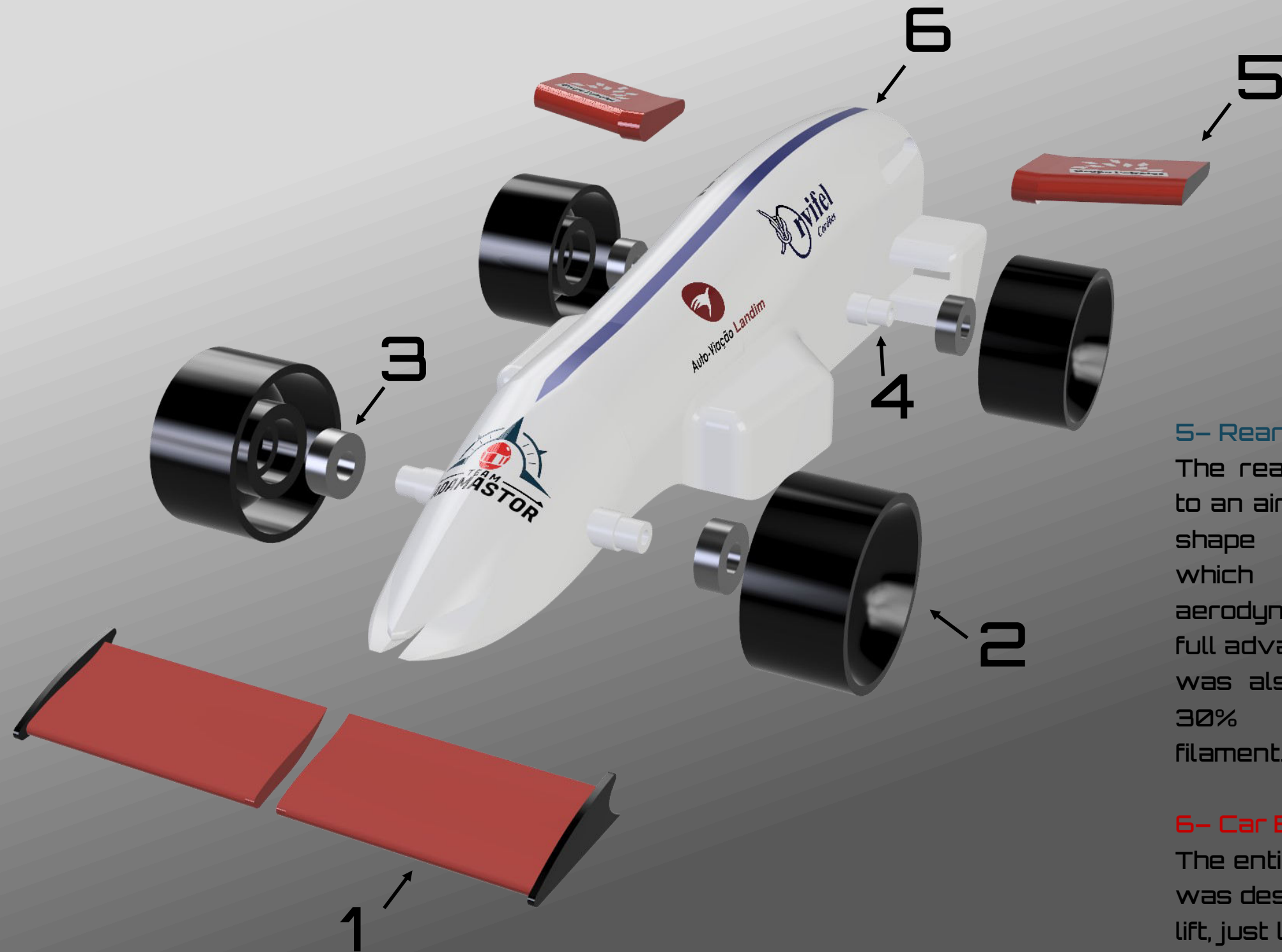
Wheel material is not yet defined. However we intend these to be light and grippy.

3-Bearings:

The bearings being used are ceramic bearings, as they have less friction and, in turn, less temperature variation

4-Axes:

The axes are made of aluminum as it is an easy-to-handle, light and resistant material.



Label

5- Rear Airfoil:

The rear airfoil is similar to an airplane wing (in the shape of a teardrop, which helps the aerodynamics a lot) taking full advantage of the lift. It was also printed with a 30% carbon fiber filament.

6- Car Body:

The entire body of the car was designed to generate lift, just like an airplane. Its aerodynamics generate as little friction as possible.

